

Sudharsan Nadar

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Profile Summary

Computer Engineering graduate from **SIESGST Nerul** with hands-on experience building intelligent, data-driven systems across **ML, backend development, and cloud infrastructure**. Proficient in **Python, FastAPI, Flask, SQL, and AWS**, with published IEEE research in deep learning and a strong project record spanning **NLP pipelines** and full-stack platforms. Actively seeking roles in ML engineering, data analytics, agentic AI, or Python development.

Technical Skills

Languages : Python, PHP, C/C++, JavaScript, TypeScript

Backend : REST APIs, Fast API, Flask, PyTorch, TensorFlow, Pandas, scikit-learn, NLTK, Levenshtein, Librosa, NumPy, HuggingFace

Databases : PostgreSQL, MySQL

Frontend : React.js, Flutter, Bootstrap

Tools & DevOps : Git, Postman, AWS, XAMPP, Figma, Android Studio

Education

SIESGST Nerul

Aug 2022 - Present

B.E in Computer Engineering

Expected Graduation: May 2026

Relevant Coursework: Data Structures and Algorithms, Operating Systems, DBMS, CN, OOP, Software Engineering

CGPA : 7.5/10

Experience

EasyGoLife Pvt. Ltd | **Web Developer Intern**

Remote | Feb 2025 - May 2025

- **Key Skills:** PHP, SQL, Front-End Development, HTML, CSS, Bootstrap
- Led a team of three interns to build a **full-stack** society management platform.
- Contributed to frontend design, backend logic using PHP, and **database structuring** with MySQL.
- Managed task allocation ensured timely delivery, and **integrated key modules** including complaints, maintenance, visitor approvals, and member management.

SIES GST | **Software Developer**

Remote | July 2025

- **Key Skills:** PHP, JSON, Front-End Development, XAMPP, PhpMyAdmin
- Developed an **intelligent FAQ chatbot** for the college admission portal using PHP with similarity and Levenshtein-based matching.
- Built a **mobile-responsive interface** and integrated it seamlessly into the website footer.
- Improved user experience by delivering accurate, instant, **link-supported responses** for admission-related queries.

IFuture Technologies Pvt. Ltd | **Cloud Engineer Intern**

On Site | Jan 2025

- **Key Skills:** Cloud Computing, Microsoft Server Management
- Completed a cloud-focused internship centered on **Microsoft Server Management** and **private cloud environments**.
- Gained hands-on experience in managing cloud infrastructure, **configuring servers**, and deploying offline applications.
- Recognized for strong **problem-solving, consistency**, and professional discipline.

Projects

CropSure: Crop Damage Assessment & Agricultural Insurance Platform— [GitHub](#)

- Developed an end-to-end **deep learning** system using **NASNet** and **Vision Transformer** architectures to detect crop damage from multi-source satellite imagery (Sentinel-1, Sentinel-2, Landsat-8) via Google Earth Engine.
- Engineered **NDVI** and **SAR** feature extraction pipelines for disaster detection and automated compensation estimation.
- Built a **TypeScript + Python full-stack** platform enabling farmers to submit insurance claims digitally, reducing manual assessment overhead and supporting scalable, data-driven agricultural insurance verification.
- Built an **Flutter** based **Mobile Application** for mapping the farmer land using the **StreetMaps API** and optimized **SQL Schema** for storing the live Data.
- Optimized the Backend for **API connection** between the App and the **Backend**.

CHSMitra: Society Administration platform – [GitHub](#)

- **Engineered a full-stack society management and maintenance billing platform**, streamlining resident operations by automating 50–60% of manual workflows across billing, complaints, visitor approvals, and member records.
- **Implemented secure billing logic with automated invoice generation and Razorpay payment integration**, reducing payment delays and enabling real-time transaction verification with low-latency callbacks.
- **Optimized MySQL queries and database schema**, improving data retrieval speeds by ~35% and enhancing stability for multi-module access.
- **Developed modular PHP backend components** for CRUD operations, role-based access, and session-secured actions, increasing maintainability and reducing code redundancy.
- **Designed responsive UI flows** using HTML, CSS, Bootstrap, and JavaScript, improving user interaction smoothness and mobile compatibility.

Aura Voice: Automatic Pronunciation Error Detector – [GitHub](#)

- **Developed an AI-based pronunciation evaluation system** that analyzes user speech, compares it with correct phoneme patterns, and provides real-time corrective feedback for English learning.

- **Implemented speech preprocessing** and feature extraction using *Librosa*, *SoundFile*, *NumPy*, and *SciPy*, enabling accurate detection of pitch, MFCCs, and acoustic deviations. **Processed audio inputs with SpeechRecognition.**
- **Used NLTK and Levenshtein distance algorithms** to perform string similarity, phonetic comparison, and error scoring for mispronounced words.
- **Built RESTful APIs with Flask 3.1**, integrated CORS handling, and used *Flask-SQLAlchemy* for database operations and structured audio feedback history.
- **Integrated with Flutter frontend**, for audio upload, error visualization, and interactive pronunciation correction for learners.

Aathichudi Tamil phrase learning application – [GitHub](#)

- **Developed a Python full-stack** pronunciation learning platform enabling users to view Tamil phrases, listen to reference audio, record their own pronunciation, and receive instant correctness feedback.
- Built a **full-stack pronunciation** evaluation system with a **React frontend (Web Audio API, Axios, modular components)** and a **FastAPI backend** supporting audio uploads, real-time scoring, and structured **REST responses**.
- Implemented audio preprocessing and analysis **pipelines using Pydub, ffmpeg-python, SpeechRecognition, and Levenshtein distance** to normalize audio, transcribe speech, and compute phonetic similarity.
- **Optimized backend** performance with **Uvicorn** workers, enabling low-latency feedback and a seamless “correct / try again” learning flow across the **integrated React–FastAPI architecture**.

Attendance Tracker using AWS Cloud & Cognito - [GitHub](#)

- **Developed backend logic** using **PHP** to handle subject creation, attendance marking, percentage calculations, and validation flows with secure **session management**.
- **Designed** and structured the **database using MySQL**, implementing relational tables for users, subjects, and attendance records with **optimized queries** for fast read/write operations.
- **Integrated AWS Cognito authentication**, using hosted UI and token verification to enable secure login, protect routes, and maintain user-level data isolation.
- **Implemented dynamic attendance dashboards** using **JavaScript** to generate real-time percentage charts and progress summaries for each subject. **Built a fully responsive frontend** using **HTML, CSS, and Bootstrap** to streamline user interaction.

Weather & Air Quality App– [GitHub](#)

- **Developed** a cross-platform **mobile application** using **Flutter**, enabling users to fetch real-time weather data and AQI details based on **GPS** coordinates or selected cities.
- **Integrated OpenWeatherMap APIs** for temperature, humidity, pollution index, and particulate data, with robust **JSON parsing and error-handled network calls** and trigger **dynamic API requests** for accurate environmental metrics.
- Built a **reactive UI using Flutter** widgets, displaying AQI labels (Good, Moderate, Poor, etc.) with dynamic theme updates driven by air quality and weather conditions a structured application **logic in Dart** with clean state management.
- Optimized app performance with efficient **API handling**, caching strategies, and resource-aware widget rendering for smooth user experience across devices.

College Admission FAQ Chatbot– [SIES | GST](#) / [GitHub](#)

- Designed and implemented a **rule-based** FAQ chatbot using **PHP**, leveraging **string similarity and Levenshtein distance** algorithms to accurately match user queries with predefined answers.
- Structured and optimized **JSON-based** knowledge storage to ensure fast lookup, easy updates, and consistent response formatting.
- Developed **backend logic** to dynamically return **contextual responses, hyperlinks, and structured text**, improving query resolution accuracy.
- Integrated the chatbot into the website footer using **HTML, CSS, and JavaScript**, ensuring **mobile responsiveness and cross-browser compatibility**

ASSESSMENTS / CERTIFICATIONS

AWS- Cloud Academy

Successfully completed a comprehensive **AWS Cloud Academy** course hands-on experience in **core cloud services such as EC2, S3, Lambda, and IAM**.

Research & Publications

AuraVoice – Automatic Pronunciation Error Detection and Correction Application

Peer-reviewed work detailing the design of an **AI-based pronunciation** error detection system that analyses user speech using MFCC extraction, phoneme comparison, and Levenshtein distance scoring to deliver real-time corrective feedback for English learners.

Cropsure: A Deep Learning Framework for Crop Damage Assessment and Agricultural Insurance Support

Published research investigating the application of NASNet and Vision Transformer architectures for agricultural crop loss prediction, enabling **data-driven** insurance support for farmers. Demonstrated comparative model performance across **deep learning** frameworks with real- world impact potential.

Leadership Quality

Senior Logistics Coordinator – CSI (Computer Society of India)

- **Managed logistics** for **technical events, workshops, and seminars** by coordinating with Teammates, allocating resources, and overseeing on-ground operations. Ensured smooth event execution and contributed to the efficiency and professionalism of CSI’s student-led initiatives.